Module Information Pack

CSY2087 Data Structures and Algorithms

The University of Northampton

Contents

Introduction

This Module Information Pack provides information about the module to enable you to deliver it in a way that is comparable to that at the University of Northampton and at the University's other partners. Comparability across modes and locations of delivery is a key part of the quality assurance and enhancement process.

The focus of this document is the module. Throughout the document you will find references to further information that is provided through your Partner Resource Centre. It is important that you read this document in conjunction with the information in your Partner Resource Centre.

If, after reading this document and the information in your Partner Resource Centre, you have any questions about the module, please contact the University's Module Leader. You can find their name and contact details in your Partner Resource Centre.

Module Specification

The module specification can be obtained through your Partner Resource Centre. The module specification is the definitive description of the module purpose, content, and assessment strategy, and is the basis from which module delivery is planned and materials prepared.

The specification is provided in the form of a link to the definitive version of the document held by the University's Curriculum Team. It is important that you refer to the definitive version for any changes that may have been made prior to delivery.

Sample Teaching Schedule

A sample teaching schedule is provided as an appendix to this document to help you plan the structure of your teaching. You are free to use this schedule, or to adapt it. If you choose to adapt it, please ensure that the topics you cover remain aligned with the module's learning outcomes, the indicative content, and the requirements of the assessment items (found in the module specification). Please also ensure that the order in which topics are delivered remains aligned with the assessment schedule.

Assessment

This module is assessed through **two** items of assessment. These are described later in this section. You must use the assessment briefs provided by the University.

The actual assessment briefs for use in the current academic year will be provided through your Partner Resource Centre. Assessment items that are distributed to students later in the module may not be provided to you until after module delivery has commenced. Examination papers are provided in a different way – please read the 'examinations' section later in this section.

Assessment Deadlines

Each assessment brief includes four deadlines that must be adhered to;

Submission deadline – this is the date by which students must submit their work. Where work is submitted electronically, the deadline is 2359hrs (UK time) on the specified date.

Grade return – this is the date by which you must make moderated grades and feedback available to students. In accordance with University policy, this is a maximum of four working weeks from the submission deadline.

Resit deadline – this is the date by which students needing to complete the resit assignment must submit their work. Where work is submitted electronically, the deadline is 2359hrs (UK time) on the specified date.

Resit grade return – this is the date by which you must make moderated resit grades and feedback available to students. In accordance with University policy, this is a maximum of four working weeks from the resit submission deadline.

If any of the assessment briefs provided do not contain all of these dates, please ask the University's Module Leader. You must not set your own dates.

Assessment Submission

All assessments for this module use SaGE (Submitting and Grading Electronically). This means that students must submit their work electronically through NILE.

NILE provides module leaders with a choice of tools for assignment submission. The description of each assessment item (below) identifies which submission tool is to be used. Using the same tool across the modes and locations of delivery supports the provision of comparable feedback, and also enables the External Examiner to complete their role more effectively.

Guidance on SaGE is provided through your Partner Resource Centre.

Assessment Grading and Feedback

Each assessment brief includes a set of assessment criteria, which are derived from the University's generic assessment criteria for the relevant academic level. Student work is assessed against the stated criteria, and feedback should reflect strengths and weaknesses of the work against these.

NILE provides module leaders with a choice of tools to grade work and to provide feedback to students. The description of each assessment item (below) identifies which grading/feedback tool is to be used. Using the same tool across the modes and locations of delivery supports the provision of comparable feedback, and also enables the External Examiner to complete their role more effectively.

Guidance on SaGE is provided through your Partner Resource Centre.

Assessment One

Assessment one is a Time Constrained lab (TCL) which will be conducted during normal lecture time. The actual date and time have not been available yet.

The TCA is consisted of multiple question covering contents delivered, and it runs within a 2 hours session.

Assessment Two

Assessment two is a Time Constrained lab (TCL) which will be conducted during normal lecture time. The actual date and time have not been available yet.

The TCA is consisted of multiple question covering contents delivered, and it runs within a 2 hours session.

Students submit this assessment using the TurnitIn tool. When setting up the TurnitIn submission point, we use the default settings. We provide feedback using a rubric and the Grading feedback section on NILE.

Sample teaching materials are provided within the Partner Resource Centre. We provide these to illustrate the style, standard, volume and level of material that we use for delivery at the University. We do not provide all of our teaching materials because a) we know that teaching from someone else's materials is very difficult, b) we believe it is important that you are able to use your own examples, case studies, anecdotes, and preferred learning resources, c) we want to encourage and enable you to take responsibility for delivery of the module at your location.

Sample Teaching Materials

Sample teaching materials are provided as an appendix to this document. We provide these to illustrate the style, standard, volume and level of material that we use for delivery at the University. We do not provide all of our teaching materials because a) we know that teaching from someone else's materials is very difficult, b) we believe it is important that you are able to use your own examples, case studies, anecdotes, and preferred learning resources, c) we want to encourage and enable you to take responsibility for delivery of the module at your location.

Sample Reading List

A sample reading list can be obtained from your Partner Resource Centre. This is provided in the form of a link to the reading list that is in use at the University.

You are welcome to use the contents of the sample list, to adapt it, or to develop your own. You should <u>not</u> direct students to the sample list because a) you will not have any control over the content of the list, b) the Module Leader at the University may annotate the list with notes that are specific to students at the University, and c) it is possible that some resources on the list may not be available to your students.

In developing your list, you must ensure that it only includes resources that are available to your students.

Appendix One – Sample Teaching Schedule

Please, note: from the academic year 2021/2022 the teaching of all UON 20 credits modules will change from a yearlong 26 weeks to a 14 weeks Semester. However, the delivery content will remain the same. That will normally, mean the delivery of two weeks material in one week.

Week	Topics
1	**Module introduction and overview**
	Introduction to HTML **Activity**
2	**Control Structures, Loops, Strings, Random Numbers**
	Functions
3	**Arrays**
	Searching and Sorting Algorithms, Vectors
4	**Pointers**
	String and C Strings
5	**Structures**
	Recursion
6	**Revision for TCA 1**
	Revision for TCA 1
7 AW1	**TCL 1**
8	**Object Oriented Programming**
	Friends, Operator Overloading and more on objects

Week	Topics
9	**Inheritance**
	Polymorphism
10	**Exceptions, templates, STL and vectors**
	Linked Lists
11	**Stacks and Queues**
	Binary trees
12	**Expression Evaluation**
	Graphs
13	**Files**
	Revision for TCA 2
14 AW2	**TCL 2**